

Final Project Jake

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4/24/26

Introduction

My individual research question is whether a house's value as an asset drives the actual median value of house prices in the Boston area. Using strictly "asset" features (e.g., property tax rate), my experiment contrasts my teammates who are predicting median house value using two other categories of predictors: i) socioeconomic/environmental factors and ii) locational factors.

The data set is sourced from the ISLR2 R package's Boston dataset which, itself, is sourced and modified from the MASS R package.

Here, I will regress median house value onto a subset of the available features which we have manually grouped as "asset value" predictors: property tax rate per \$10,000, average number of rooms per home, and age. Age is not the median/average age of a housing unit. Instead, age is the proportion of housing units built before 1940.

To answer the question of whether a house's asset value drives its actual value I will evaluate three regression approaches using root mean squared error (RMSE) as the evaluation metric: i) ridge regression, ii) random forests, and iii) natural splines.

Data

```
library(ISLR2)
library(tidyverse)
```

Warning: package 'ggplot2' was built under R version 4.5.2

Warning: package 'tibble' was built under R version 4.5.2

Warning: package 'tidyr' was built under R version 4.5.2

Warning: package 'readr' was built under R version 4.5.2

Warning: package 'purrr' was built under R version 4.5.2

Warning: package 'dplyr' was built under R version 4.5.3

Warning: package 'stringr' was built under R version 4.5.2

Warning: package 'lubridate' was built under R version 4.5.2

-- Attaching core tidyverse packages ----- tidyverse 2.0.0 --

v dplyr 1.2.1 v readr 2.2.0

v forcats 1.0.1 v stringr 1.6.0

v ggplot2 4.0.2 v tibble 3.3.1

v lubridate 1.9.5 v tidyr 1.3.2

v purrr 1.2.1

-- Conflicts ----- tidyverse_conflicts() --

x dplyr::filter() masks stats::filter()

x dplyr::lag() masks stats::lag()

i Use the conflicted package (<<http://conflicted.r-lib.org/>>) to force all conflicts to become

```
library(glmnet)
```

Loading required package: Matrix

Warning: package 'Matrix' was built under R version 4.5.3

Attaching package: 'Matrix'

The following objects are masked from 'package:tidyr':

expand, pack, unpack

Loaded glmnet 4.1-10

```
df_bos <- as_tibble(Boston)
df_bos <- df_bos |> select(tax, rm, age, medv)
head(df_bos)
```

```
# A tibble: 6 x 4
   tax    rm  age medv
<dbl> <dbl> <dbl> <dbl>
1   296  6.58  65.2   24
2   242  6.42  78.9  21.6
3   242  7.18  61.1  34.7
4   222  7.00  45.8  33.4
5   222  7.15  54.2  36.2
6   222  6.43  58.7  28.7
```

```
summary(df_bos)
```

tax	rm	age	medv
Min. :187.0	Min. :3.561	Min. : 2.90	Min. : 5.00
1st Qu.:279.0	1st Qu.:5.886	1st Qu.: 45.02	1st Qu.:17.02
Median :330.0	Median :6.208	Median : 77.50	Median :21.20
Mean :408.2	Mean :6.285	Mean : 68.57	Mean :22.53
3rd Qu.:666.0	3rd Qu.:6.623	3rd Qu.: 94.08	3rd Qu.:25.00
Max. :711.0	Max. :8.780	Max. :100.00	Max. :50.00

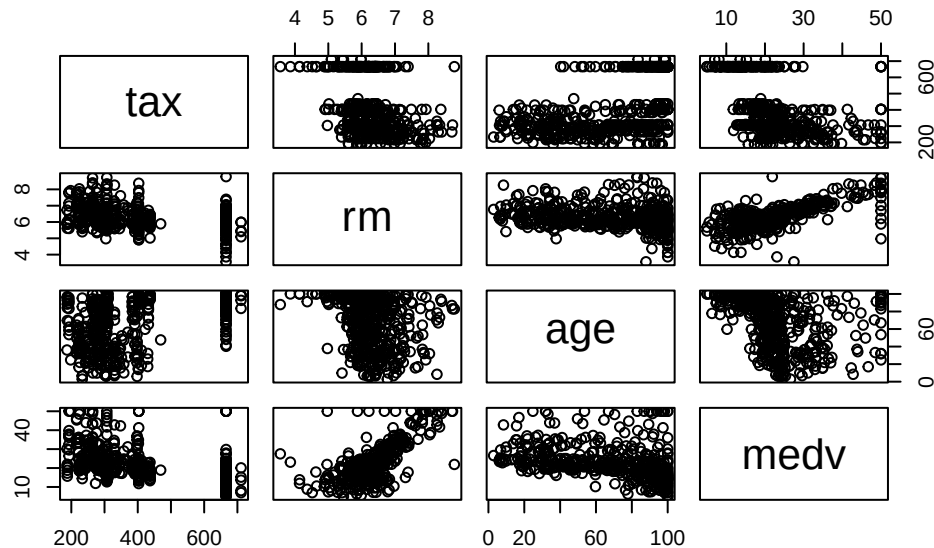
```
# check na values
colSums(is.na(df_bos))
```

```
tax    rm  age medv
  0     0    0    0
```

```
# check for duplicates
sum(duplicated(df_bos))
```

```
[1] 0
```

```
pairs(df_bos)
```



Qualitatively, there are associations between median value and the predictors, especially number of rooms.

For evaluation, I will hold out 20% of the data for testing and use the remaining 80% for cross-validation to find optimal hyperparameters for each model.

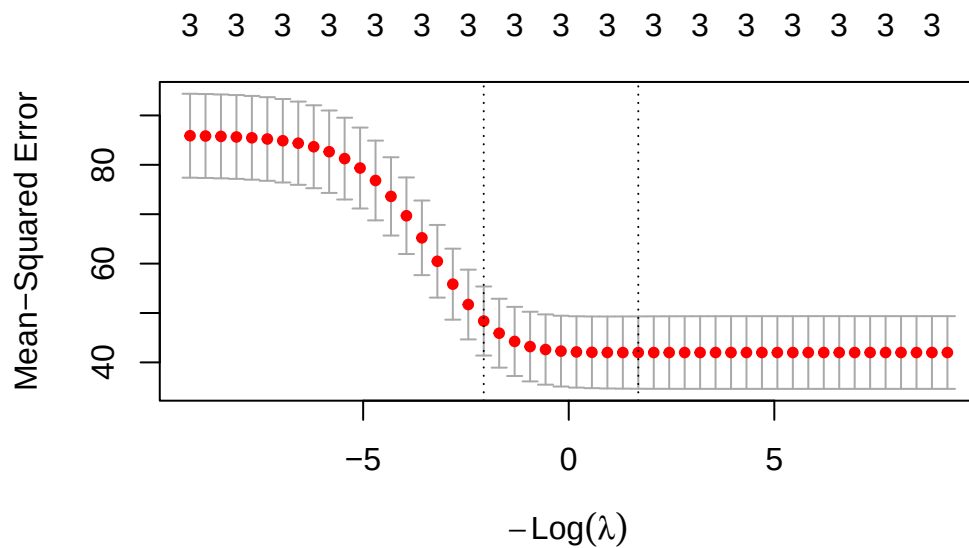
```
set.seed(0)
train_index <- sample(1:nrow(df_bos), size = 0.8 * nrow(df_bos))
train_data <- df_bos[train_index, ]
test_data <- df_bos[-train_index, ]
```

Models

Ridge Regression

```
# glmnet requires matrices as input
# and does not use formula syntax
x_train <- as.matrix(train_data |> select(-medv))
y_train <- train_data$medv
x_test <- as.matrix(test_data |> select(-medv))
y_test <- test_data$medv
```

```
# lambdas ranging from 10^4 to 10^-4
lambdas <- 10^seq(4,-4, length = 50)
# alpha=0 is ridge
alpha <- 0
cv_ridge <- cv.glmnet(x_train, y_train, alpha = alpha, lambda = lambdas)
ridge_best_lambda <- cv_ridge$lambda.1se
plot(cv_ridge)
```



The best lambda is 7.9060432.